

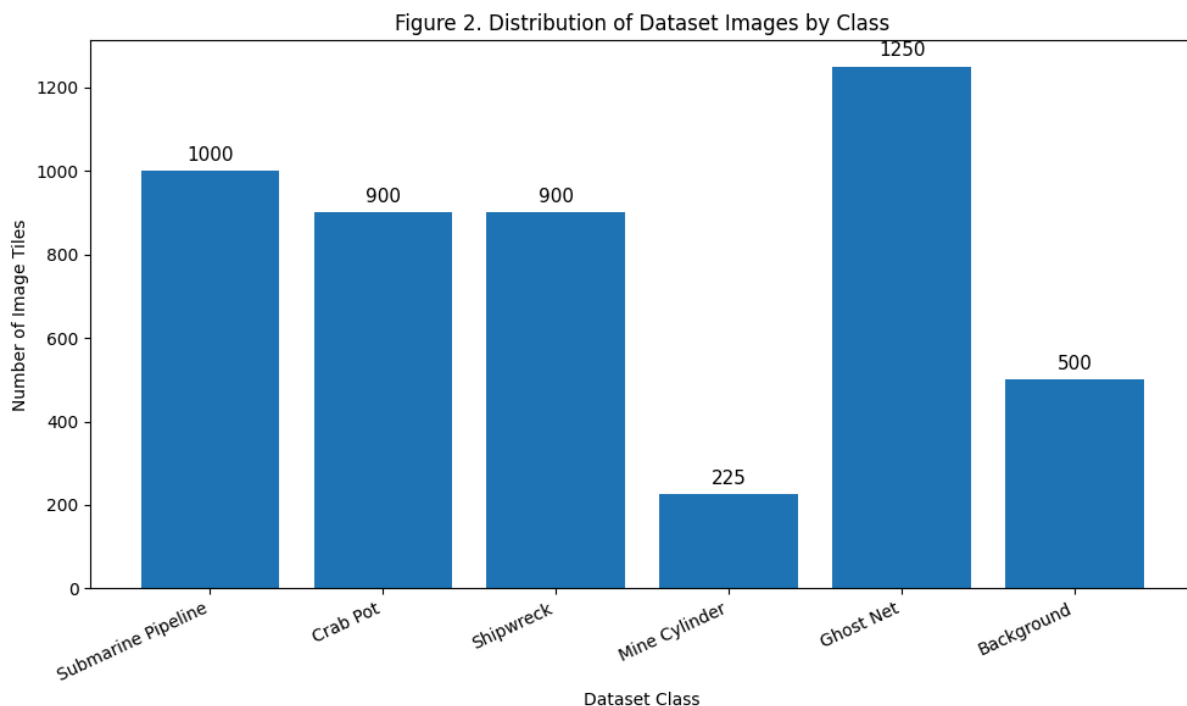
Data and preprocessing

1. Data Collection

Public labelled SSS data exists exactly for one class that is crab pots.

Target Class	Public Data situation	What we did
Shipwrecks	AI4Shipwrecks provides 286 labelled sonar images from 28 shipwreck sites.	We divided the sonar data into tiles and added some ship images for training
Pipes	There is no large dedicated public pipe dataset.SubpipeMini2 provides some SSS data	We divided available strips into 1000 smaller traing images.
Cylinders	Very little data	We used some sonar mine data from kaggle and added some synthetic data to increase training data
Ghost Nets	NO public dataset available	We created synthetic ghost-nets SSS.

The complete dataset contained **6,405 tiles**, including training, validation and test data.Around 74% of the training data came from real survey data, while around 26% was synthetic.Synthetic data was mainly used where suitable real data was not available.



If the model is trained only on images containing objects, it may start detecting normal seabed structures as debris.To reduce this problem, we included approximately 500 object-free background tiles. These are called hard negatives .They teach the model that an empty seabed is also a valid image and should not automatically produce a detection.This is important for reducing false positives.

This created a problem because one complete wreck could become one very large bounding box. As a result, only about 152 usable tiles were produced from the first processing method. We changed the approach and divided the strips using 50% overlap. A morphological closing operation was also used so that broken parts of a shipwreck could be treated as one object. This showed that improving the way the data was prepared was more useful than simply changing model settings.

Crab-pot data was initially included as a trained class because labelled public data was available. However, its detection performance remained low despite increasing the amount of training data, indicating that the main problem was class separability rather than simply dataset size. The class was therefore removed from the final product. In addition, the corresponding data is excluded from the public dataset release due to licensing restrictions.

Dataset Licensing

Each external dataset is used according to its respective licence. Datasets with academic-only or non-redistributable terms are not included in the public dataset release. The trained model and internal dataset should therefore be distinguished from the publicly distributable dataset.

2. Pre-Processing

Side-scan sonar images are different from normal photographs. Preprocessing is used before the image is given to the detection model.

The complete preprocessing design contains 6 stages:

1. Water-column removal: - When the sonar is above the seabed, the sonar signal first travels through water before reaching the bottom. This produces an empty region in the sonar image. If this region is given directly to the model, it can sometimes create false detections. This stage mainly helps to improve precision by removing an area where false detections can occur.

2. Slant-to-ground range conversion: - A sonar does not directly measure the horizontal distance to an object. It measures the slant range, which is the diagonal distance between the sonar and the target. Because of this, objects at different distances can appear with different scales in the image. To correct this, slant range can be converted to ground range. $\text{Ground Range} = \sqrt{(\text{Slant Range}^2 - \text{Height}^2)}$. After this conversion, the image can be resampled to a more uniform spatial resolution.

3. Radiometric and time-varied gain correction: - Sonar intensity becomes weaker as the signal travels farther through water. Because of this, objects close to the sonar may appear brighter while objects farther away may appear darker. This can make object detection difficult. A gain correction can be applied to reduce this effect. The idea is to compensate for the loss of acoustic energy with distance so that an object does not become difficult to detect simply because it is farther from the sonar. This correction is expected to help mainly with recall, especially for objects located at the far end of the sonar swath.

4. Motion and attitude compensation: - For an underwater vehicle, changes in pitch, roll and heave. The sonar data can then be adjusted according to the vehicle's movement. Lost or corrupted sonar pings can also be handled using neighbouring data where possible. This stage mainly helps the shape and position of detected objects line up correctly

5. Speckle filtering: - Speckle is one of the common types of noise found in sonar images. A simple blur is not a good solution because it can remove useful information along with the noise. For DRISHTI, we use an edge-preserving Lee filter. The Lee filter looks at a small neighbourhood around each pixel. It calculates the local mean and variance and then decides how much smoothing is required. The basic equation is: $\hat{I} = \mu + W(I - \mu)$ Where:
 I = observed pixel intensity

μ = local mean intensity calculated over the 7×7 neighbourhood
 W = Lee filter weighting factor calculated from local image variance
 and estimated noise variance

The Lee filter combines the observed pixel value with the local mean according to the estimated amount of speckle noise.

6.CLAHE contrast enhancement:-The second preprocessing operation used in the shipped pipeline is CLAHE, or Contrast Limited Adaptive Histogram Equalization. Normal histogram equalization works on the complete image. For sonar images, this can sometimes increase the noise in flat areas. The image is divided into small regions and contrast is adjusted separately. In our implementation: Grid size: 8×8 and Clip value: 3.0. The clipping prevents very large changes in contrast. This can make weak object returns and acoustic shadows easier to see without excessively increasing noise in uniform seabed regions.

We also compared the model with and without preprocessing. The comparison used the same data and evaluation procedure. The reported precision was approximately **0.734** in both cases, and the false-positive rate was approximately **0.266**. This means that preprocessing did not provide a large direct improvement in the detector's main accuracy metrics. The main reason was that the enhanced images made acoustic shadows easier to separate, which is useful for the later confidence and geometry checking stage. So preprocessing is not included simply because it sounds useful, it is included because it supports the complete sonar processing pipeline.

Status column.

Stage	Purpose	Status
Water-column removal	Remove irrelevant water region	Designed / future
Slant → ground range	Correct geometric distortion	Designed / future
Radiometric + TVG	Correct intensity variation	Designed / future
Motion/attitude compensation	Correct platform movement	Designed / future
Lee filter	Reduce speckle	Implemented
CLAHE	Improve local contrast	Implemented

2.1. CLAHE and Lambertian Physics Model

There are two possible approaches for correcting sonar intensity.

1.CLAHE:-CLAHE is mainly an image-processing method.It is useful as it is simple,fast,requires only the image,easy to run on an edge device,does not require additional vehicle information.

2.Lambertian Physics Model:-The Lambertian approach is more physics-based.It uses information such as:sonar image,vehicle altitude,Grazing angle,Incidence geometry and It tries to separate the effects of Object reflectivity object elevation acoustic path loss.

Feature	CLAHE	Lambertian Physics Model
Basic approach	Improves the sonar image using image processing.	Uses sonar and vehicle information to understand the physical signal.
How it works	Improves brightness and contrast in small parts of the image.	Separates the sonar signal into object reflectivity,height and signal loss with distance.
Output	A clearer 2D black and white sonar image	Separates the sonar signal into object reflectivity height and signal loss with distance
When it needs	Only the sonar image	Sonar image and vehicle height and sonar angle information.
Main advantage	Fast simple and does not need extra sensor data.	More physically meaningful and can work better across many survey distances and angles.

The Lambertian method can provide a more physically consistent correction, but it requires more information and is more complicated to implement.For the current DRISHTI implementation, CLAHE is used as the simpler real-time preprocessing method.The deeper physics-based correction can be used later when the required telemetry is available.

2.2. AUV and Tow-Fish Preprocessing

The preprocessing requirements are slightly different for an **AUV** and a **towed sonar fish**.An AUV has its own onboard navigation and motion sensors.A tow-fish is connected to a surface vessel using a cable, so additional problems occur.

These include:

1 Layback Correction:-A tow-fish can be hundreds of metres behind the vessel.The GPS position of the vessel is therefore not necessarily the actual position of the sonar.Layback correction estimates the horizontal distance between the vessel and the sonar fish.This can be calculated using:Cable length Fish depth Vessel speed or measured using USBL positioning.The corrected position can then be added to the sonar data.

2 Speed Correction:-The vessel does not always travel at exactly the same speed.If the ping rate remains constant while the vessel speed changes, the sonar image can become compressed

or stretched. GPS speed-over-ground can be used to correct this. The sonar rows can then be resampled to maintain a more realistic spatial scale.

3 Swell and Heave Correction:-Surface waves can move the vessel up and down. This movement can also affect the tow-fish. The resulting sonar image may contain a wave-like distortion. Fish depth information can be used to estimate this movement and correct the sonar data. This helps create a more stable seabed image.

4 Multipath Filtering:-In shallow water, sonar energy can reflect from the sea surface or vessel wake. These reflections may appear as horizontal bands in the image. Some of these bands can look like objects and cause false detections. Spatial filtering and thresholding can be used to remove these unwanted returns.

5 Dynamic Bottom Tracking:-The distance between a tow-fish and the seabed can change during a survey. Because of this, using one fixed threshold for the seabed boundary is not always reliable. A dynamic tracking method can follow the seabed boundary across multiple pings. This helps remove temporary disturbances such as cable movement or fish schools before the sonar image is cropped.

Note:-All 5 need to be designed or implemented in future.